

Homework (Carolina Reaper)

- Q1.** Factor the expression $x^2 - 16$
(A) $(x + 4)(x - 4)$
(B) $x(x - 4)$
(C) $x(x + 4)$
(D) $x^2 - 4x + 16$
(E) $x^2 + 4x - 16$
- Q2.** A painter is creating a canvas with a length that is 2 meters longer than its width. If the area is 20 square meters, what is the difference of squares for the dimensions?
(A) $2^2 - 4^2$
(B) $4^2 - 2^2$
(C) $3^2 - 5^2$
(D) $6^2 - 8^2$
(E) $10^2 - 12^2$
- Q3.** A musician is setting up a rhythm pattern. If the frequency of the rhythm is $3x$ and the anti-frequency is $5x$, what is the difference of squares for these frequencies?
(A) $25x^2 - 9x^2$
(B) $9x^2 - 25x^2$
(C) $4x^2 - 16x^2$
(D) $16x^2 - 4x^2$
(E) $10x^2 - 20x^2$
- Q4.** Factor the expression $49y^2 - 9$
(A) $7y - 3)(7y + 3)$
(B) $7y(7y - 3)$
(C) $7y(7y + 3)$
(D) $49y^2 + 3$
(E) $49y^2 - 6$
- Q5.** A sculptor is designing a square sculpture with a side length $2x + 3$. If a smaller square with side length $x - 1$ is cut out, what is the difference of squares for the areas?
(A) $(2x + 3)^2 - (x - 1)^2$
(B) $(x - 1)^2 - (2x + 3)^2$
(C) $(3x)^2 - (x - 1)^2$
(D) $(x + 2)^2 - (2x - 1)^2$
(E) $(2x - 5)^2 - (3x + 2)^2$
- Q6.** Factor the expression $x^2 - 2x - 15$ is not in the form of a difference of squares, but $x^2 - 4x + 4 - 19$ can be factored as a difference of squares. What is the factored form?
(A) $(x - 2)^2 - 19$
(B) $(x + 2)^2 - 19$
(C) $(x - 5)(x + 3)$
(D) $(x^2 - 4x + 4) - 19$
(E) $(x^2 + 4x + 4) - 19$
- Q7.** A guitarist is using two amps with frequencies of 10 and 6. What is the difference of squares for the volumes if the volume is directly proportional to the square of the frequency?
(A) $100 - 36$
(B) $10^2 + 6^2$
(C) $10 + 6$
(D) $100 + 36$
(E) $100 \cdot 36$
- Q8.** A photographer is setting up a backdrop that is $x^2 - 4$ meters long. If $x = 3$, what is the difference of squares for the dimensions?
(A) $5^2 - 1^2$
(B) $4^2 - 2^2$
(C) $3^2 - 2^2$
(D) $x^2 - 2^2$
(E) $x^2 + 2^2$

Open Ended Questions (FRQ)

- Q9.** A sound engineer is mixing two audio signals with frequencies of $15x$ and $9x$. If the volume of the mixed signal is directly proportional to the difference of squares of these frequencies, find the difference of squares and express it as a factored expression.
- Q10.** A painter is creating a series of square canvases with side lengths $x + 2$, x , and $x - 2$. If the total area of the three canvases is $3x^2 + 12x + 12$ square meters, find the difference of squares for the areas of the first two canvases, and express it as a factored expression.

Chef's Tips

- Emphasize that the difference of squares can be applied to various real-world contexts, such as art and music.
- Make sure students factor expressions correctly and apply the pattern to solve problems.
- Encourage students to think critically when approaching word problems.